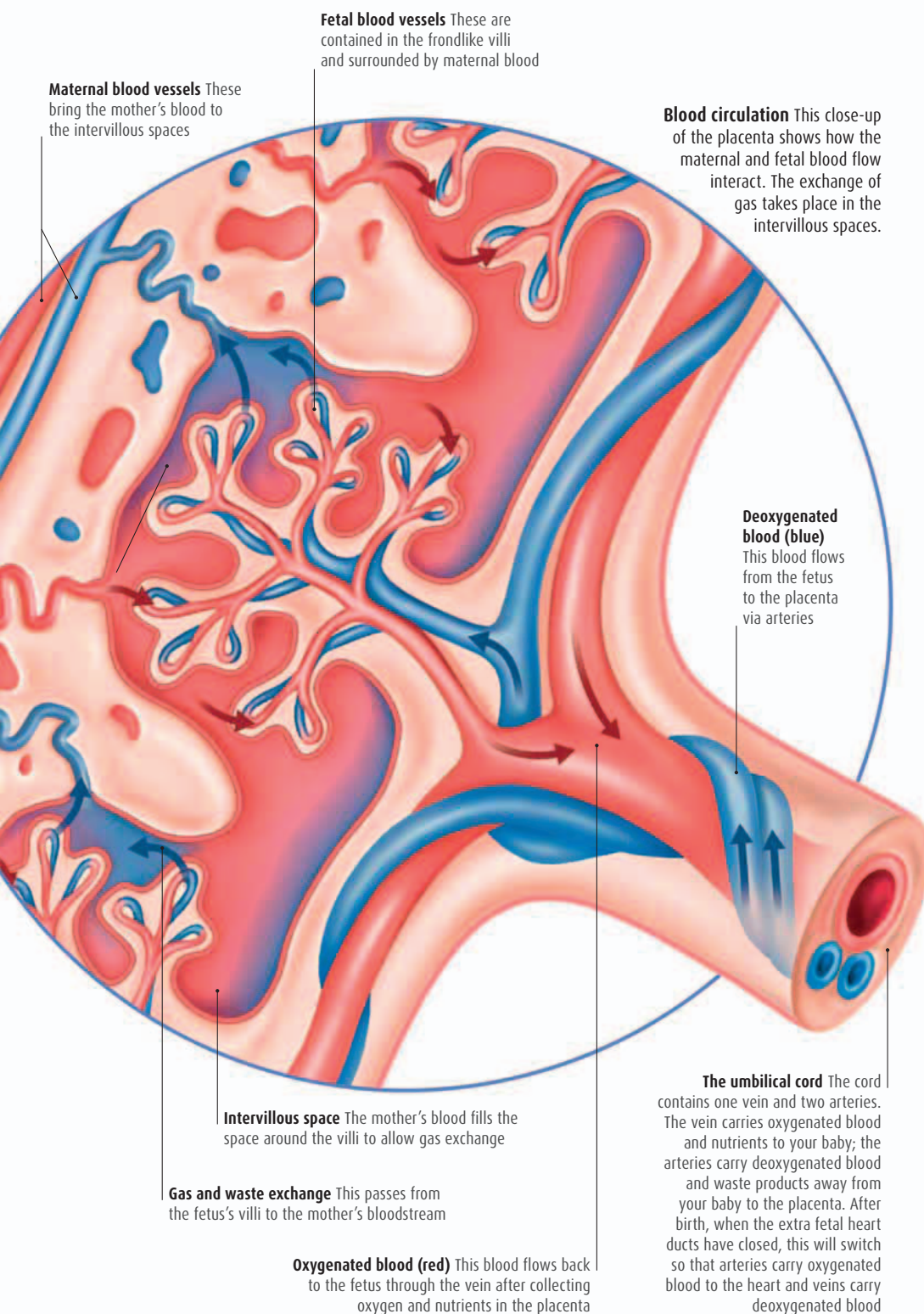




THE PLACENTA'S ROLE

The placenta plays a vital role in your baby's survival. It has four key functions.

» **Removing your baby's waste:** there are two arteries in the umbilical cord as well as arterial vessels in the villi. These carry waste and carbon dioxide-filled blood from the baby. The waste gases travel through the walls of the villi into the mother's blood and are then processed as waste by the mother's liver and kidneys.

» **Protecting your baby from disease:** the cells of the villi are tightly packed. This means that organisms that are made up of large cells, such as bacteria and viruses that might be in the mother's blood, can't pass into the baby's blood through the villi walls. Not all organisms are too large though, which is why it's important to avoid contracting certain diseases (such as rubella or chicken pox) by being vaccinated against them. Toward the end of your pregnancy, antibodies from your blood will pass into your baby's circulation; these provide her with a basic level of immunity that lasts several weeks after she's born.

» **Protecting your baby from chemicals:** many chemicals (including toxins) are also made up of cells too large to cross the placenta. This is why you can continue taking certain medications while pregnant. Certain toxins, though, such as nicotine and alcohol, can pass through the villi walls.

» **Producing hormones:** the placenta serves as its own endocrine system. It's an important source of progesterone to help maintain your pregnancy and prevent premature labor, and of estrogen, to ensure your uterus expands to accommodate your growing baby and prepare your body for labor and, later, for breast-feeding. The placenta is also the source of the hormone hCG, which is the first signal your body receives that pregnancy has occurred. Growth hormones and relaxin (which softens the ligaments in your pelvis to make space for your baby and prepare your body for birth) are also produced in the placenta, along with several other hormones that optimize the transfer of nutrients and oxygen within the placenta itself.

The placenta expands in size from just a few cells to between **12 oz (350 g)** and **1½ lb (700 g)** in weight.